Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [12CE74]

SEVENTH SEMESTER B.E EXAMINATION, DECEMBER - 2015

CONSTRUCTION PLANNING AND MANAGEMENT

Time: 3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full Questions.

1. a. What are the objectives of construction management? Name the functions involved. **8**

b. Explain how the bar charts and milestone charts help in preparing the construction schedule. **6**

2. a. Explain the following terms: **8**

- **i.** Event
- **ii.** Activity
- **iii.** Earliest Event Time
- **iv.** Last Event Time
- **v.** Slack
- **vi.** Critical path

b. Differentiate between PERT and CPM. **6**

3. a. Show the graphs and explain how the direct indirect costs vary with project duration. **6**

b. Explain the steps involved in cost optimization w.r.t. time. **8**

4. a. What are the objectives of materials management? Explain how resource smoothing and resource levelling help in handling materials. **8**

b. What is the ABC classification of inventory? **6**

5. a. What are the factors affecting the cost of owning and operating construction equipment? **6**

b. Explain how the time and motion study helps in improving productivity. **8**

6. a. What is total quality management? **7**

b. What are the different quality assurance techniques? **7**

7. a. Explain the different approaches to improve safety in construction. **8**

b. Explain the terms: **6**

- **i.** Equity
- **ii.** Annuity
- **iii.** Benefit-cost ratio

8. a. Explain the concept of MIS. **6**

b. What is scheduling? Explain the classification of scheduling. **8**

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